

A \$12 million battery. A \$57 million project.

Hyde Park BESS (Project 21334): where the cost of a utility-owned, distribution-connected battery storage station actually sits, and how it compares to industry benchmarks

Cost Estimating Center of Excellence | Estimate basis: ESF Total Rev 3, 24 Sep 2026 | 29 Sep 2026

Internal. Contains vendor-confidential pricing.

THE PROMISE

By the end of this deck you can answer one question in one breath

“Why is a \$12 million battery a \$57 million project?” Because a vendor prices the battery. We build the station, the site, the feeder protection and the siting case, and we carry the risk and the cost of money.

\$12.1M

Vendor fixed-price contract

Nomad cover agreement, 10 Mar 2026.
12 Voyager LFP units, PCS, BMS, fire protection, 10-yr warranty. \$605/kWh, \$1,211/kW.

\$57.2M

Project 21334 approval level

ESF Total Rev 3. Station 360 \$50.0M plus Station 496 protection \$7.3M. \$2,862/kWh, \$5,725/kW. Conceptual, -25%/+50%.

21%

Battery share of the total

Add the D-Line circuit extension (24211, \$4.8M conceptual) and the share drops to 20%. Four dollars of every five are not the battery.

Three sections: where the money sits (thirteen layers, unit costs, benchmarks); how the cost changed (2021 to today); can the number be defended (red team, stress tests, what must close).

What a vendor quote contains, and what it leaves for the utility

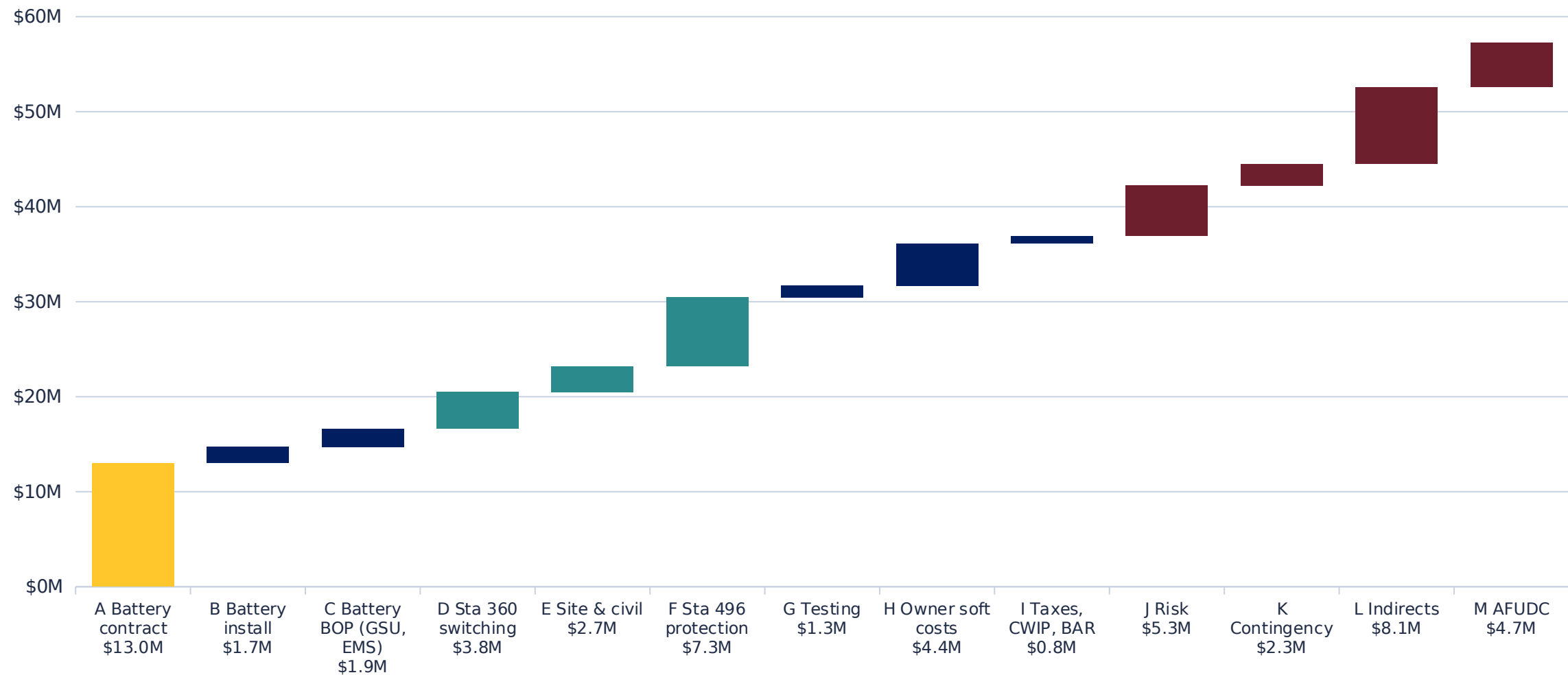
Inside the \$12.1M Nomad price

- 12 containerized LFP units, 10 MW / 20 MWh, 12 MVA
- Inverters (PCS), BMS, HVAC, fire detection and suppression inside the container
- Spares \$125K, shipping \$90K, remote support \$210K
- 10-year warranty; years 3-7 extension \$798K
- Field assembly supervision, commissioning, training: \$246K
- Fixed until Final Acceptance; LDs capped at 10%

Outside it: \$45.1M, 79% of the project

- **Station 360**, \$3.8M direct: four 15 kV breaker skids, GSUs, SSVT, metering, cable, duct bank, ground grid
- **Site**, \$2.7M direct: a 200 ft x 200 ft platform on a closed landfill, access road, retaining wall
- **Station 496**, \$7.3M: 20 relays, transfer trip, RTU work so four feeders can accept 2.5 MW each
- **Owner scope**, \$10.1M: install labor, GSUs, EMS, testing, engineering, Article 80 siting, outreach, PM, taxes
- **Station 360 risk, contingency, indirects, AFUDC**, \$20.3M: 35% of the total (Station 496 carries its own inside its \$7.3M)

Thirteen layers between the battery contract and the approval level



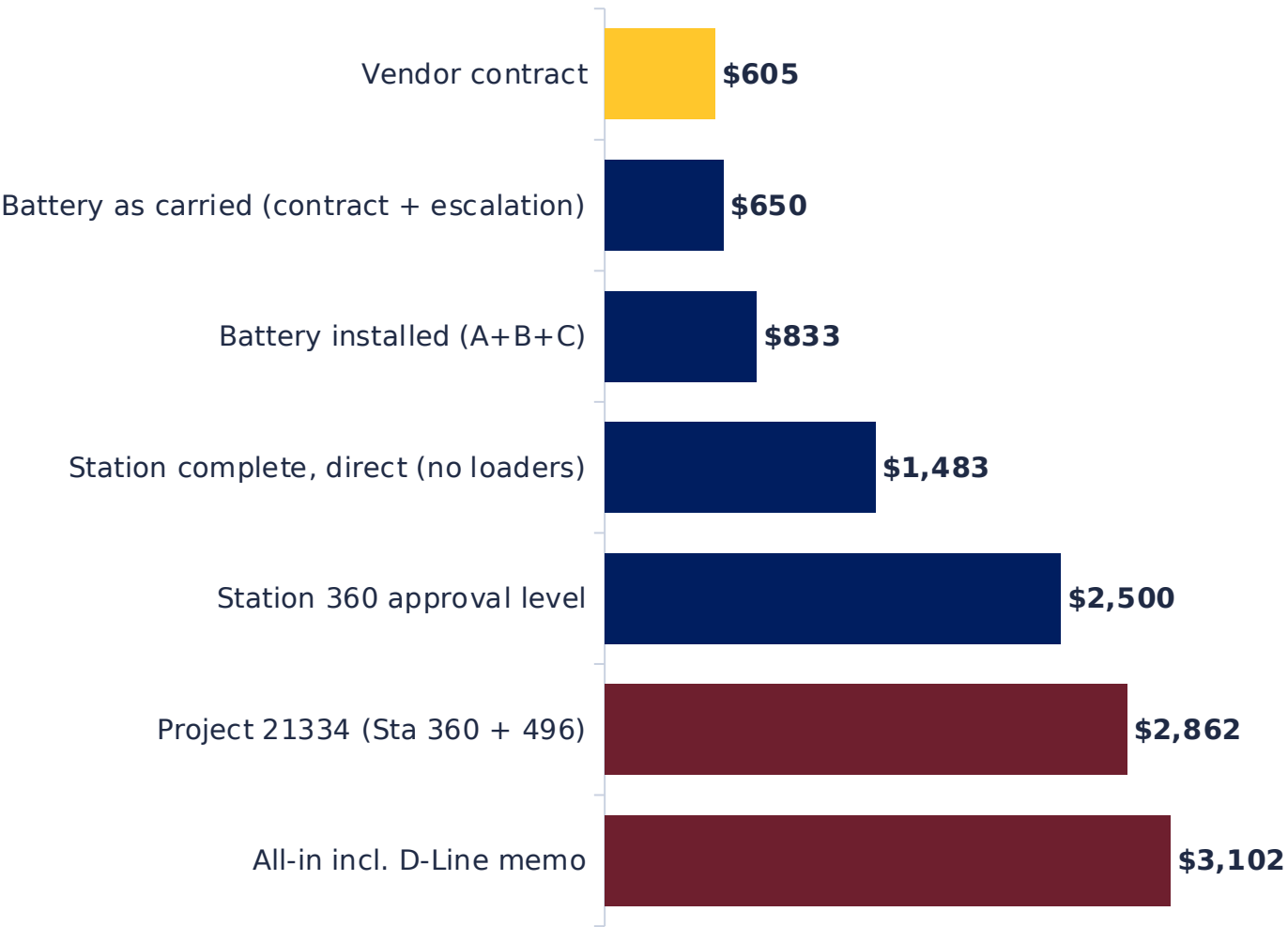
Gold = battery contract as carried in Rev 3 (\$12.1M contract plus \$1.1M escalation on the unpaid balance; see slide 16). Teal = station, site and feeder protection. Burgundy = risk, contingency, indirects, AFUDC. Total \$57.2M.
EstimatingCoE | Eversource Energy | Stewards of Affordability

Where the \$57.2M sits when risk, contingency, indirects and AFUDC are spread

Layer	Direct	Loaded	Share
Battery supply contract (Nomad)	\$13.0M	\$21.9M	38%
Battery installation & integration (owner scope)	\$1.7M	\$2.9M	5%
Battery balance-of-plant (GSUs, EMS/comms, aux power)	\$1.9M	\$3.3M	6%
Station 360 13.8 kV switching & interconnection facilities	\$3.8M	\$6.5M	11%
Site development & civil (former landfill site)	\$2.7M	\$4.5M	8%
Station 496 feeder protection upgrades (interconnection)	\$7.3M	\$7.3M	13%
Testing & commissioning	\$1.3M	\$2.1M	4%
Owner soft costs (engineering, siting, environmental, outreach, PM, land)	\$4.4M	\$7.4M	13%
Taxes, CWIP, builders-risk insurance	\$0.8M	\$1.3M	2%
Total, project 21334 (directs incl. \$20.3M of loaders J-M)	\$57.2M	\$57.2M	100%

Battery-related layers A+B+C loaded: \$28.1M (49%). Station, site and feeder interconnection D+E+F loaded: \$18.3M (32%). Owner soft costs, testing and taxes: \$10.9M.

The same project is \$605/kWh or \$2,862/kWh depending on where you draw the line

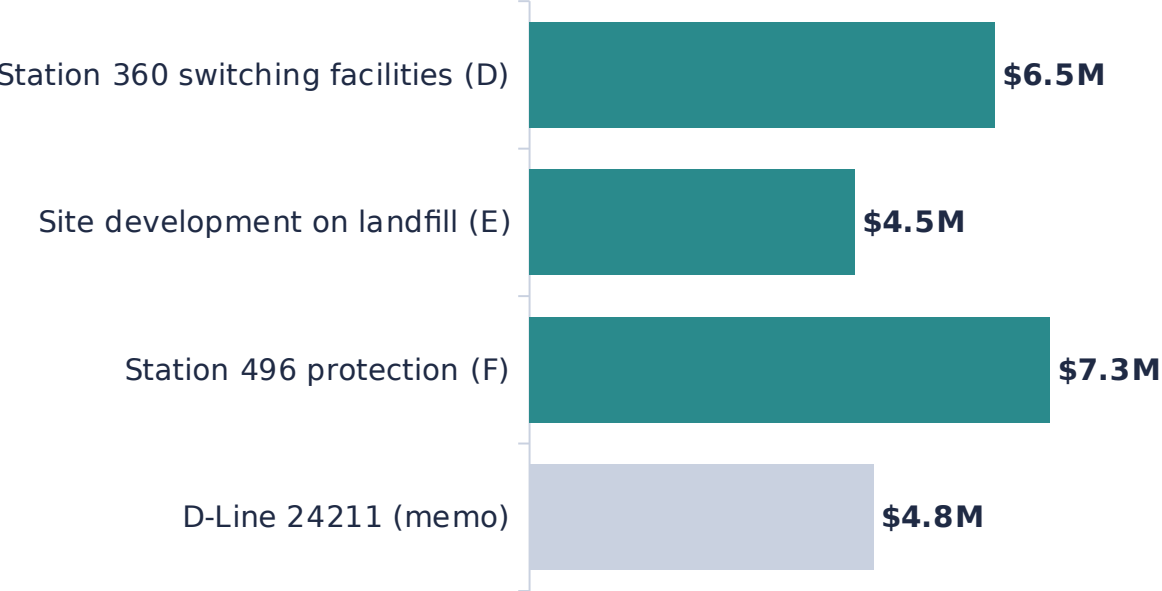


Boundary	\$/kW	\$/kWh	x
Vendor contract	\$1,211	\$605	1.0x
Battery installed	\$1,666	\$833	1.4x
Sta 360 approval	\$4,999	\$2,500	4.1x
Project 21334	\$5,725	\$2,862	4.7x

Incl. D-Line \$6,204 \$3,102 5.1x

A benchmark is only meaningful against the row that matches its definition. Pack and DC-block prices belong beside the top bar; NREL and EIA installed costs beside 'battery installed' or 'station complete'; nothing published includes a utility's own indirects and AFUDC.

Station and interconnection: \$18.3M loaded, none of it in a vendor quote

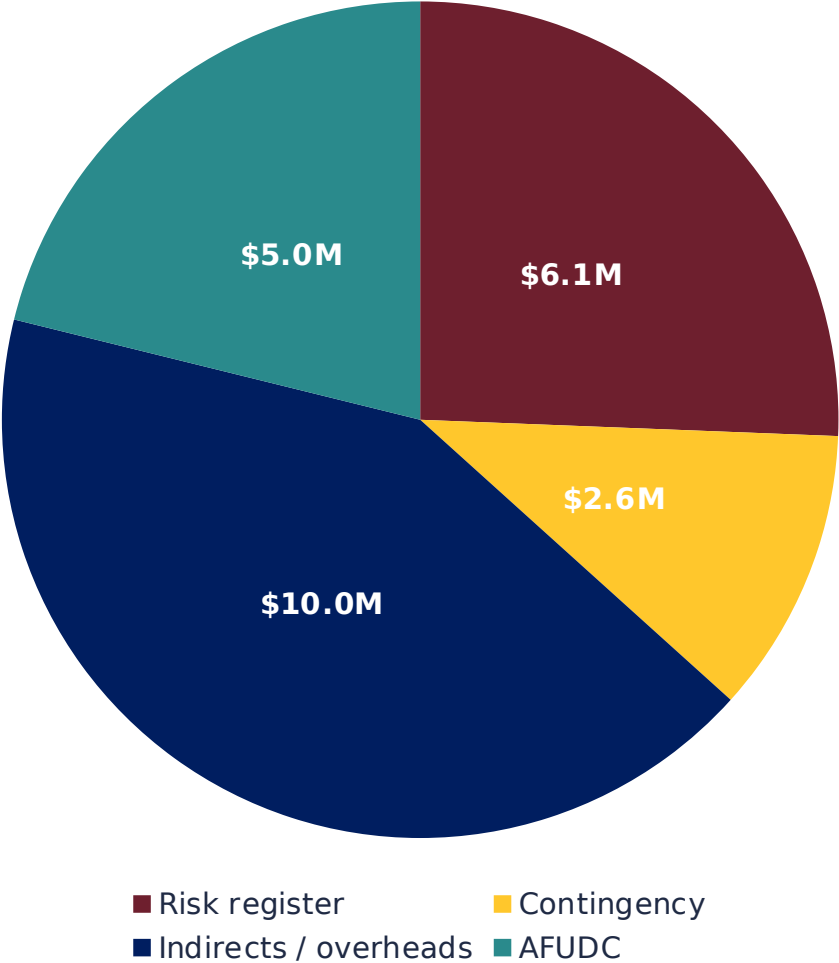


- **Four Siemens SDV7 breaker skids at \$300K each** (relays, PTs, DC battery, cabinet on the skid), disconnects, four metering transclosures, station service, riser poles, 15 kV cable and 1,240 ft of duct bank.
- **Station 496:** 20 SEL relays replacing SEL-551 and REF610 units on four feeders, RTU and RTAC points, transfer trip over dark fiber; \$1.8M of that is testing.
- **Site:** 2.5 acres cleared, 760 ft road, 2,520 CY cut, imported fill, 14 ft retaining wall, ground grid, fence, lighting; Haugland's platform take-off is larger than ours and is unreconciled.
- **D-Line (24211):** 5,465 ft feeder extension, three manholes, 16,395 ft of 3-700 cable; \$4.8M is a 2023 conceptual figure.

\$18.3M
Loaded, in 21334
32% of the approval level

\$23.1M
Incl. D-Line memo
37% of the all-in view

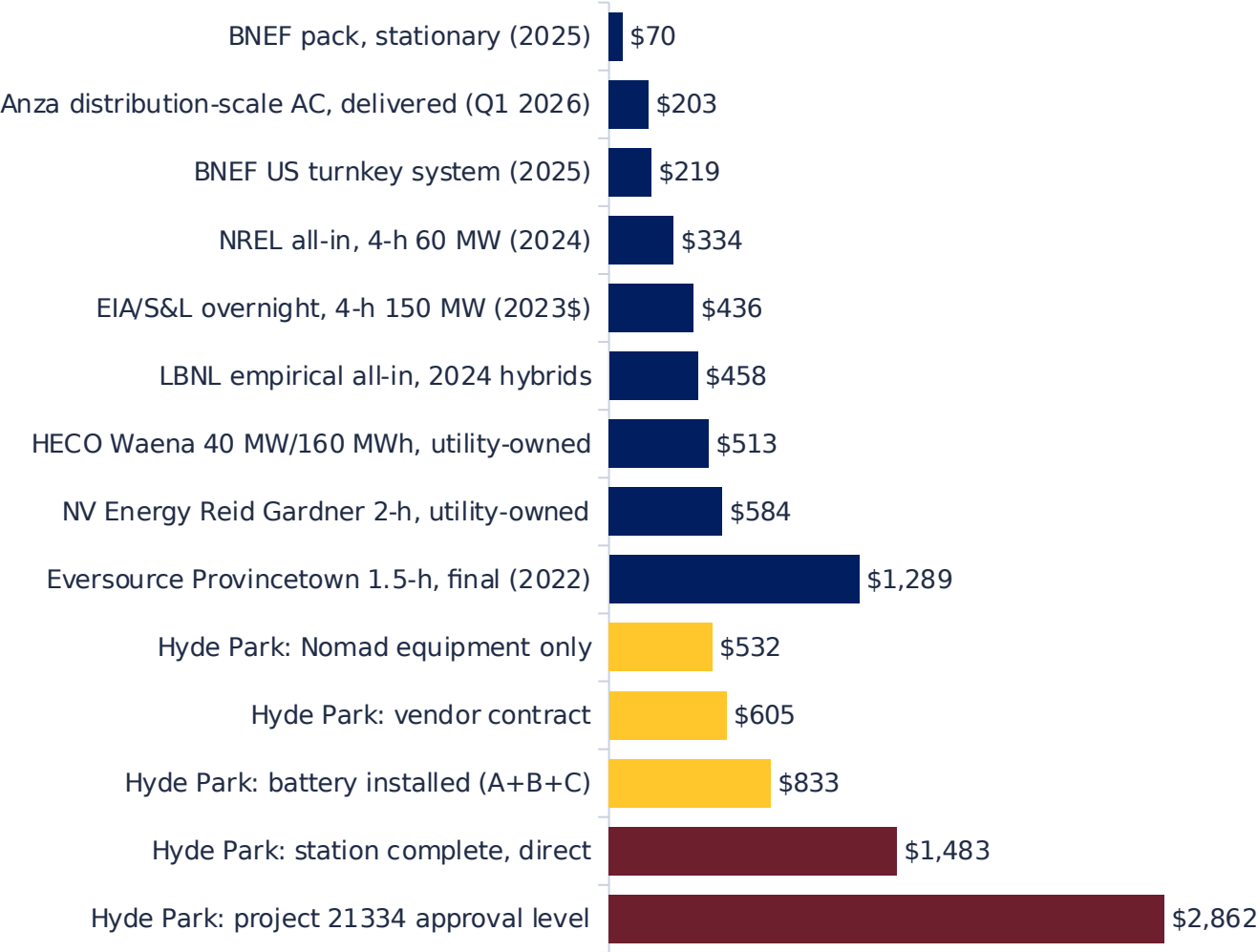
Risk, contingency, indirects and AFUDC are 41% of the approval level



Station 360 risk item	\$K
Material cost, tariffs	1,000
Design / scope change orders	800
Labor cost, shortage, overtime	800
Unforeseen UG obstructions (landfill)	450
Environmental (DEP, landfill closure docs)	300
Soil contamination remediation	300
Station and P&C engineering changes	600
Weather, siting delay, outreach, BAR deductibles	1,050

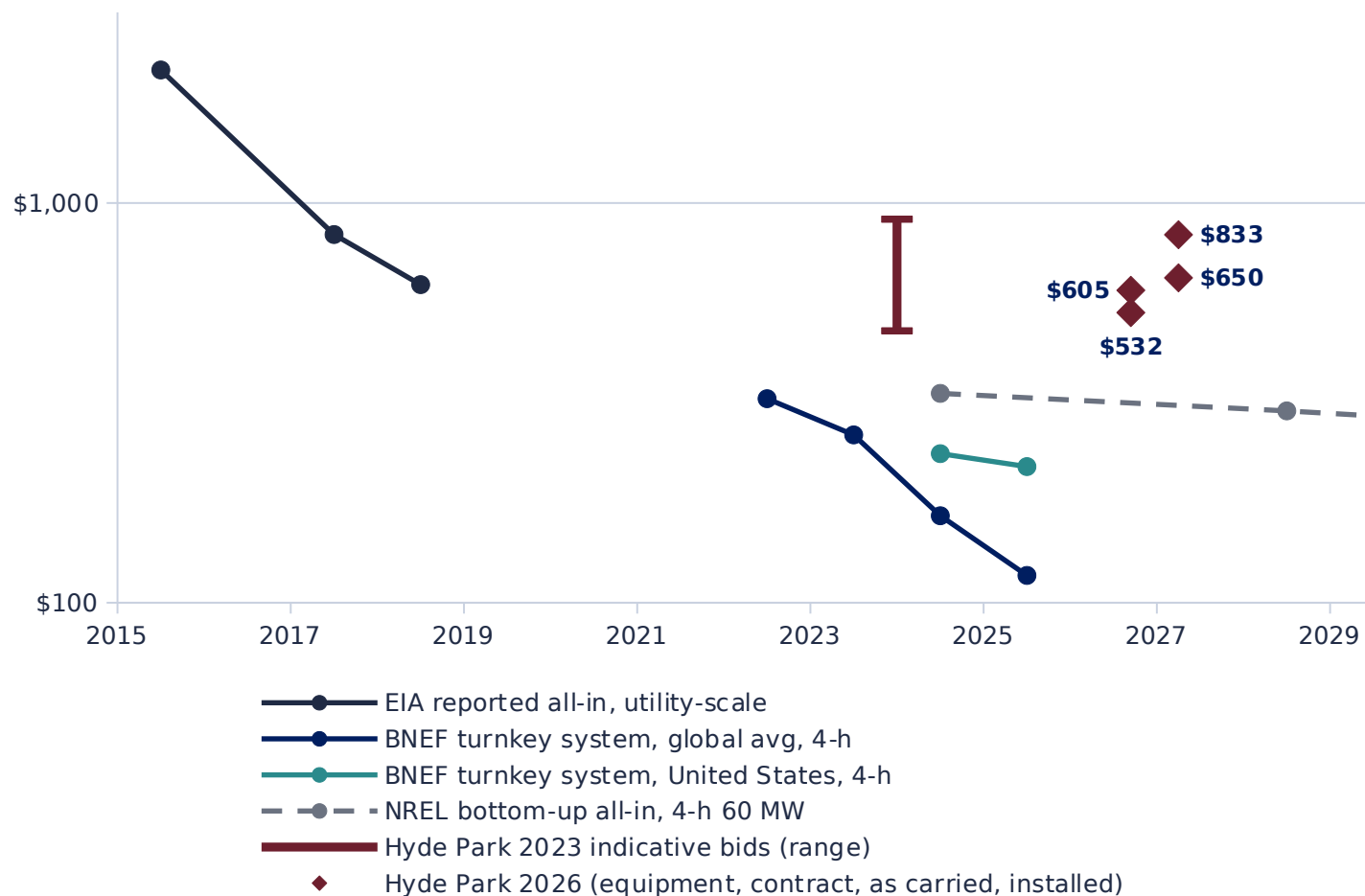
Risk plus contingency is 17.9% of the approval level before loaders, inside the 15-18% lead-review guidance. Indirects at 17.4% and AFUDC at 8.8% follow the EMA station template on a spend curve that runs from July 2021 to December 2028. Haugland's Attachment 4-F carries overlapping exposures (geotech \$0.6M-\$3.5M, fault-study equipment change \$0.25M-\$1.8M) that have not been reconciled with this register.

The battery is at market; the project is above it for reasons benchmarks exclude



- **Battery at market.** Nomad's \$532/kWh equipment price sits between BNEF's US turnkey (\$219, 4-h, 100 MW class) and the disclosed utility-owned projects (\$513-584). For a 12-unit, 2-h, FEOC-era order in New England that is a fair price.
- **Installed battery (A+B+C) at \$833/kWh** is above every national all-in index because those indices are four-hour and 60-150 MW. NREL's own cost structure (pack \$/kWh x hours + BOS \$/kW) puts a 2-h system at roughly 1.3-1.5x the 4-h \$/kWh.
- **The project at \$2,862/kWh** is not a battery benchmark at all. It carries a switching station, feeder protection at Station 496, a landfill platform, Article 80 siting, indirects and AFUDC. The closest published analogue is Provincetown at \$1,289/kWh (2022 dollars, no separate station, existing site).
- **No published index includes a utility's indirects or AFUDC.** Strip J-M and Hyde Park is \$1,483/kWh direct; strip the station and site layers D and E as well and it is \$1,156/kWh.

Battery installed-cost curve: the Nomad price sits on it once duration and size are adjusted



■ **The curve is a 4-h, 60-150 MW, real-dollar curve.** EIA's reported all-in fell from \$2,152 (2015) to \$625 (2018); BNEF turnkey from \$324 (2022) to \$117 global and \$219 US (2025); NREL's 4-h bottom-up base is \$334 (2024), \$247 by 2035.

■ **Hyde Park is 2-h, 10 MW, nominal.** NREL's structure puts a 2-h system at 1.3-1.5x the 4-h \$/kWh; the size elasticity in Appendix B adds 1.5x at 20 MWh. Adjusted US turnkey: \$440-\$505. Nomad equipment: \$532.

■ **Install is the layer the curve never shows.** Owner-scope installation, integration and balance of plant (B+C) add \$183/kWh to the \$650 battery line: \$833 installed. BNEF's install sits inside a developer's EPC; ours is Eversource scope at New England rates.

■ **The 2026 turn.** Lithium carbonate roughly doubled year on year by August 2026 and tariffs lifted delivered US prices. The March 2026 fixed price sits below where a 2027 re-procurement would land.

Plotted as published: EIA-860 nominal; BNEF real survey-year; NREL 2024\$ (2028 interpolated between the 2024 base and the 2035 mid case). 2023 bar = five indicative bids, \$479-\$911/kWh, differing scopes. Retrieved from search excerpts; verify before external use.

Escalation is already in the number: \$2.9M at template rates, \$1.1M of it on a fixed price

Where the escalation sits (Rev 3, included in the directs)

Item	\$	\$/kWh
Escalation in the approval level (Sta 360 \$2,622K + Sta 496 \$326K)	\$2,948K	\$147
on the Nomad balance, which Exhibit C fixes until Final Acceptance	\$1,130K	\$57
on station, site, feeder protection, testing and owner scope	\$1,818K	\$91
Spend curve it is applied to: actuals \$5.2M, 2026 \$13.5M, 2027 \$12.5M, 2028 \$26.1M		

Same comparison, same dollars (per kWh)

Boundary	Nominal	Ex-escalation	Benchmark brought to 2028 at 3%/yr
Vendor contract	\$605	\$605 (fixed)	BNEF US turnkey \$219 (2025) = \$239
Battery as carried (A)	\$650	\$593	NREL 4-h \$334 (2024\$) = \$376
Battery installed (A+B+C)	\$833	~\$756	NREL adjusted to 2-h, 10 MW: \$750-\$870
Station complete, direct	\$1,483	\$1,352	EIA/S&L \$436 (2023\$) = \$505; LBNL \$458 = \$516

- **Rates.** ESF template escalation, about 3% a year, on the 2026-2028 spend curve. \$2.9M is 5.1% of the approval level and 7.4% of directs. Benchmarks are 2023-2025 dollars; Rev 3 is 2026-2028 nominal, so one side must move before they are compared.
- **Decision pending.** \$1.13M sits on a price the contract fixes. Remove it or move it to the risk register with a stated rationale (open item, slide 20).
- **Brought to the same year, the installed battery is on the index.** NREL's \$334 is \$376 in 2028 dollars and \$750-\$870 after the 2-h and 10 MW adjustments; Hyde Park installed is \$833.
- **What 3% does not cover.** Lithium carbonate and tariffs moved faster than 3% in 2026. The battery is fixed by contract, so the exposure is the \$3.7M of non-battery material and the labor lines. The \$1.0M tariff item in the risk register is sized on total material, battery included.
- **Schedule.** Twelve months of slip adds about \$1.2M of escalation on the \$38.6M scheduled for 2027-2028, before AFUDC, PM and indirects (about \$5M in total).
- **Ex-escalation values** strip the station's \$131/kWh from the direct boundaries; the B+C share (~\$20/kWh) is allocated pro rata and is approximate.

Six things about Hyde Park that a 100 MW greenfield benchmark never has to pay for

Size. 10 MW / 20 MWh. A switching station, a siting case and a commissioning team cost about the same at 10 MW as at 100 MW; per kWh they are ten times heavier.

Duration. Two hours. Fixed \$/kW items (PCS, breakers, GSUs, interconnection) are spread over half the MWh of a four-hour benchmark.

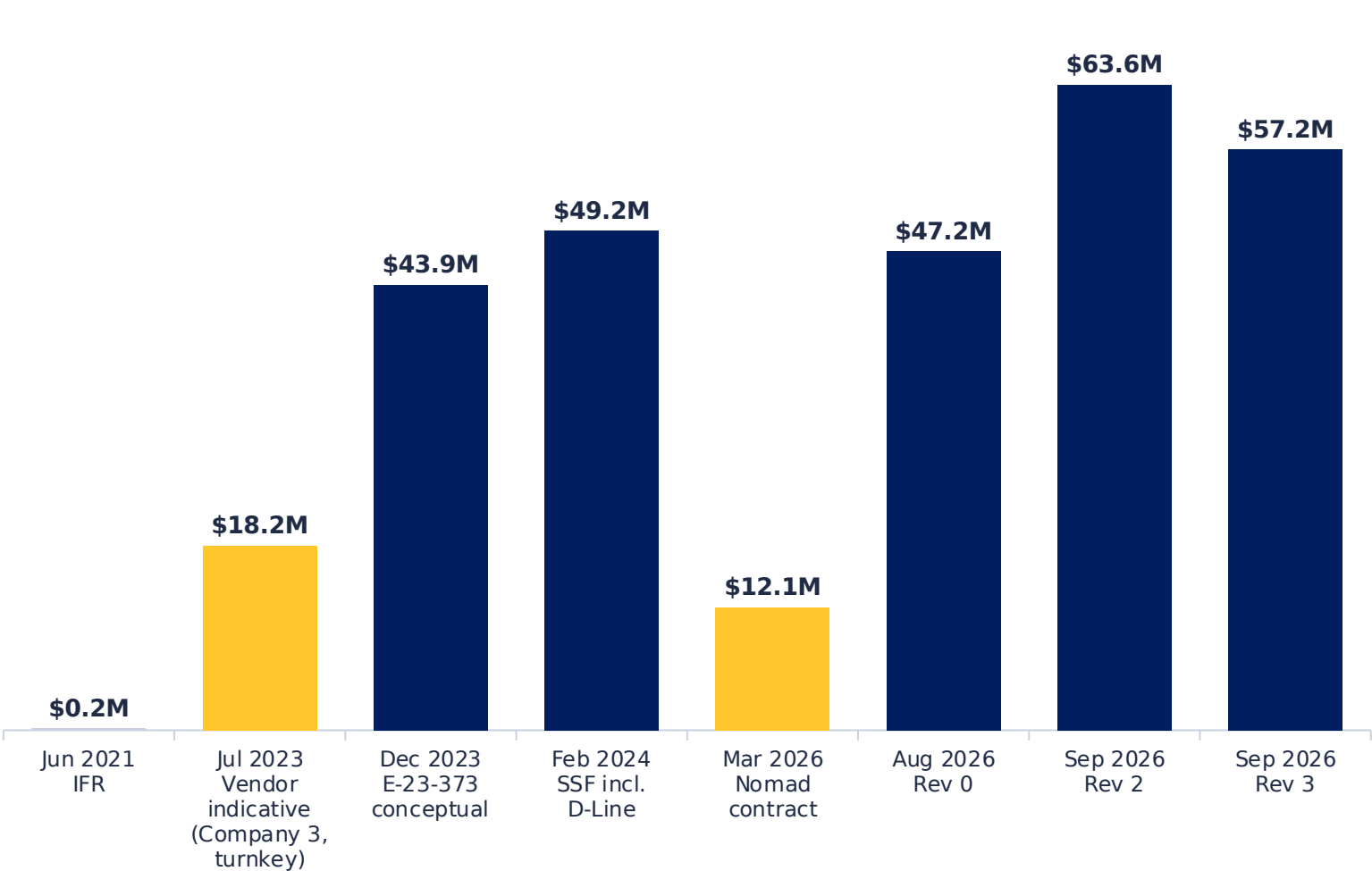
Site. A closed construction-debris landfill in Boston: DEP post-closure use permit, cap protection, 36 borings, imported platform fill, 14 ft retaining wall, unknown obstructions (\$450K risk).

Interconnection. Four feeder tie-ins at 13.8 kV, each with its own breaker skid, plus relay replacement and transfer trip at Station 496 (\$7.3M) and a 5,465 ft circuit extension (24211).

Siting and outreach. Article 80 Small Project Review, Zoning Board of Appeal, outside counsel, an EJ community, a neighborhood concerned about open land and views: \$1.5M plus \$0.5M in the risk register.

Utility accounting. Corporate indirects (\$10.0M) and AFUDC over a seven-year development window (\$5.0M) are real costs to customers that a developer's turnkey price never shows.

Five years, five in-service dates, and an estimate that moved from \$44M to \$57M

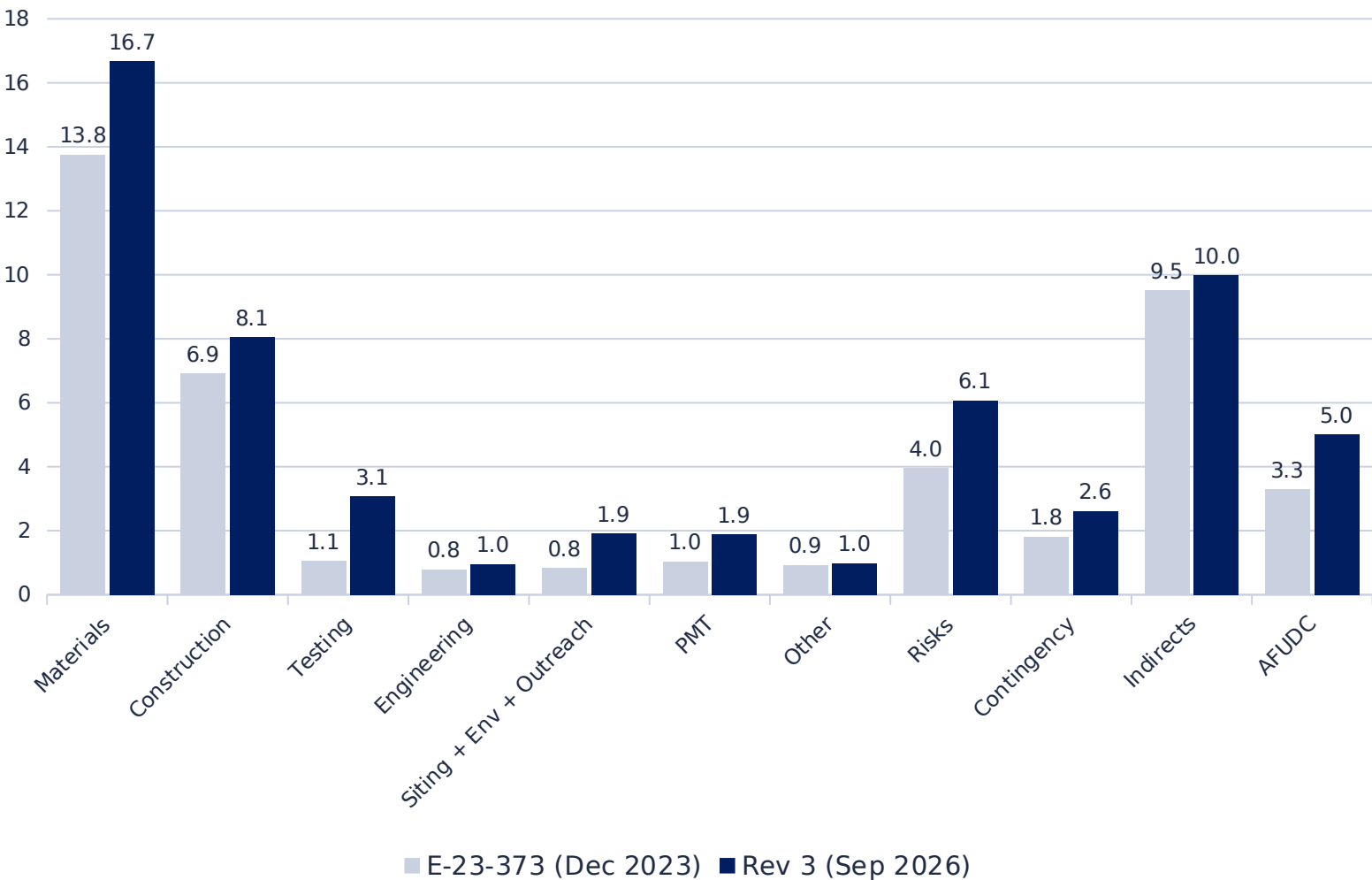


In-service date drift

As of	Planned ISD
Jun 2021	Q3 2025
Dec 2023	Apr 2027
Feb 2024	Sep 2027
Jan 2026	Aug 2028
Jun 2026	Dec 2028

Gold bars are vendor prices for the battery scope only. Navy bars are full project estimates. The battery price fell from \$18.2M turnkey-indicative (2023, incl. civil) to \$12.1M contracted (2026); the project rose \$13.3M in the same window. The 2023 conceptual carried a placeholder that was never re-based until August 2026.

What changed between the December 2023 conceptual and Rev 3: \$43.9M to \$57.2M

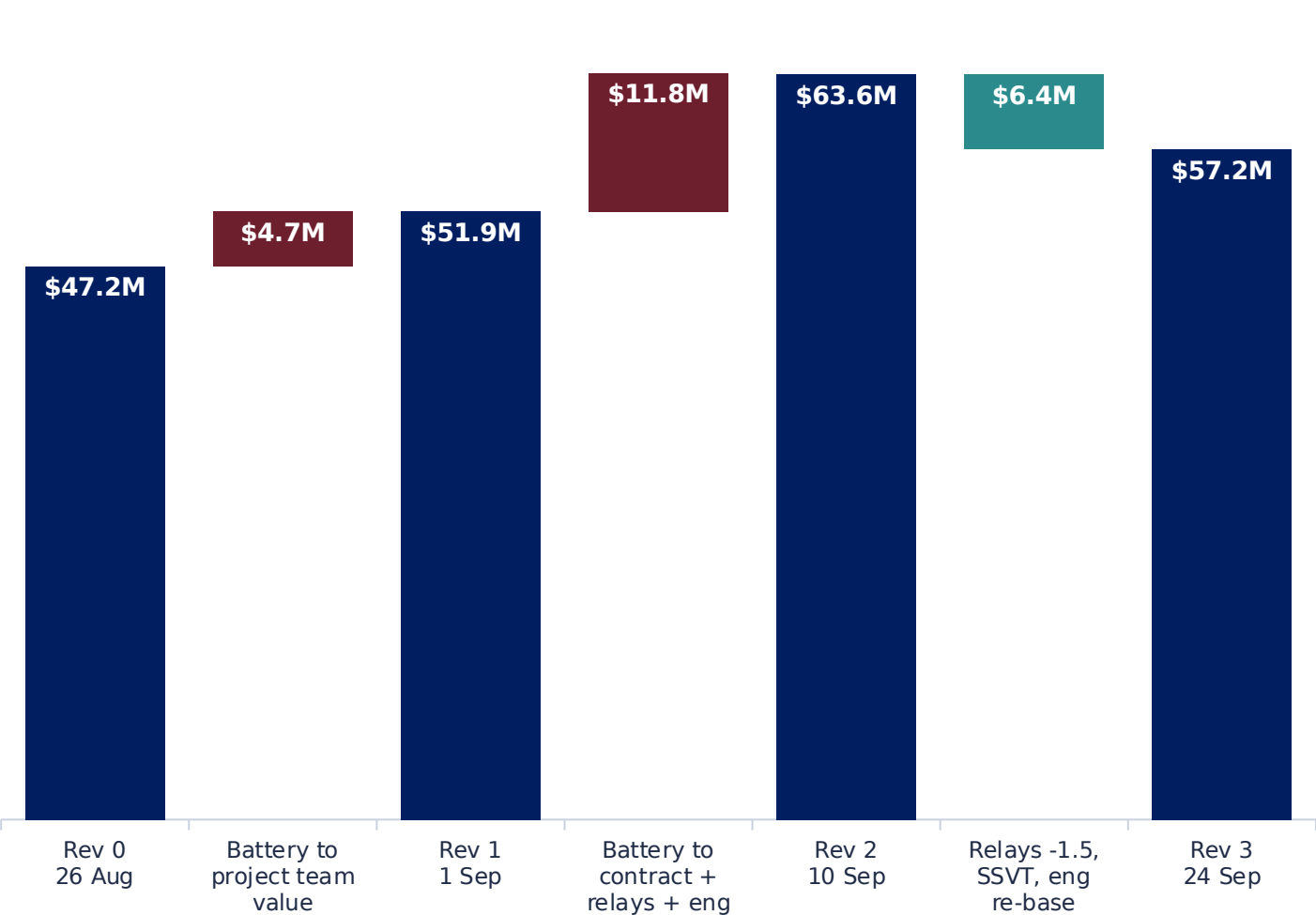


Largest movements

Category	Delta
Materials	+\$2.9M
Risks	+\$2.1M
Testing	+\$2.0M
AFUDC	+\$1.7M
Construction	+\$1.1M
Siting + Env + Outreach	+\$1.1M
PMT	+\$0.9M

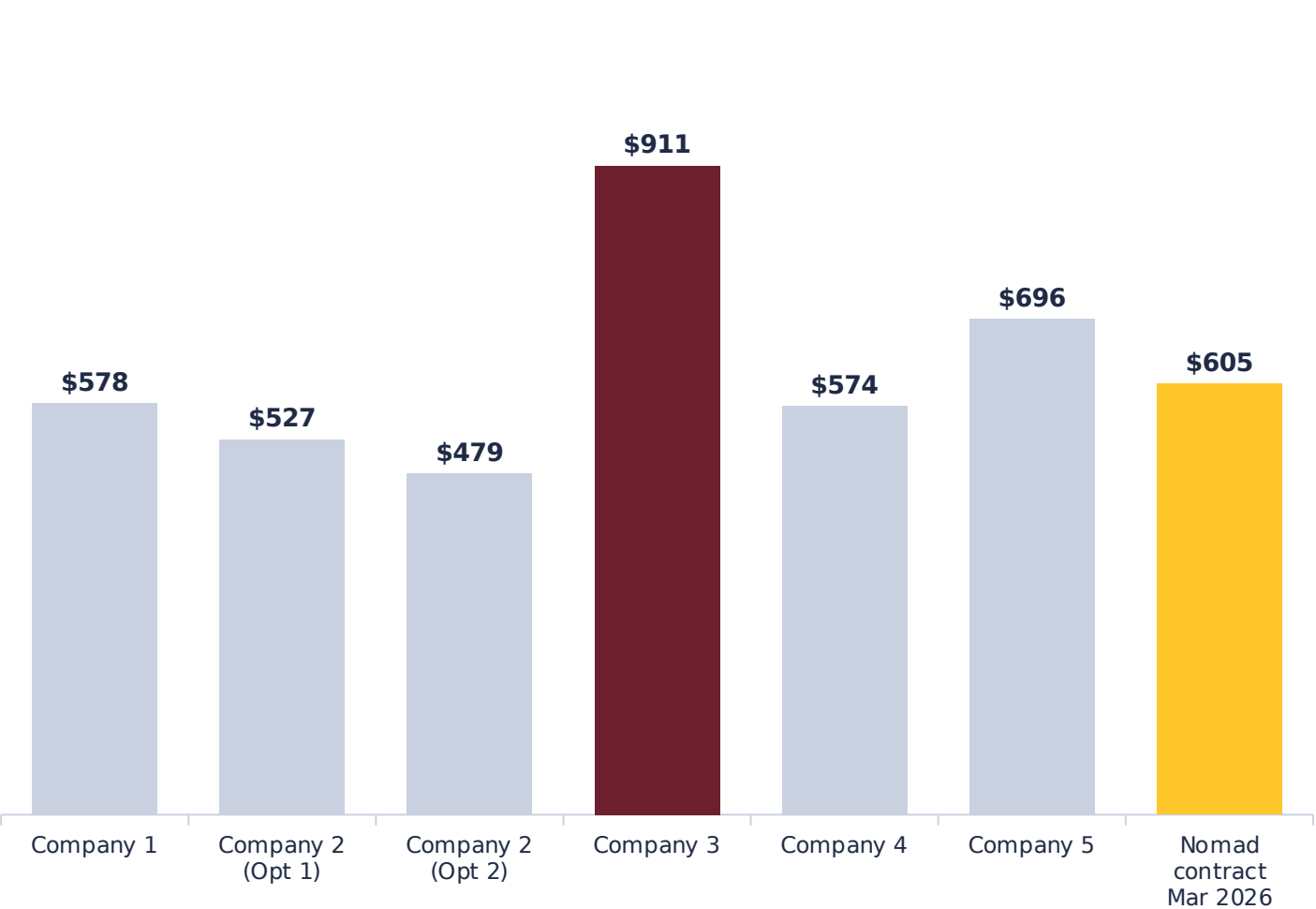
Materials: contracted battery replaces 2023 indicative pricing, plus breaker skids and GSUs. Testing and siting: Station 496 scope and Article 80 were not in E-23-373. AFUDC: a 20-month later ISD on a larger base.

Four revisions in five weeks: how the September estimate settled at \$57.2M



- **Rev 0 carried the battery at about \$6M** from the 2023 indicative sheet. The PM: 'we know the BESS system itself is 12.'
- **Rev 2 reset it to the cover agreement, escalated to \$13.1M**, and carried \$1.5M of 'relays for batteries' from a 2023 bidder that priced its own switchgear.
- **Rev 3 removed the relays** (P&C Engineering: included in the \$300K Siemens skid), cut three SSVTs to one, and re-based engineering to \$501K to go plus actuals.
- **Net:** +\$10.0M from Rev 0, of which \$7.0M is the battery line catching up to the contract.

The battery: a \$9.6M to \$18.2M spread in 2023, a \$12.1M fixed price in 2026



- **July 2023, five bidders, 10 MW / 20 MWh:** \$9.6M to \$18.2M. Every bidder excluded interconnection, permits beyond local, below-grade risk, sound walls or spares.
- **Company 3 (\$18.2M, burgundy)** became the E-23-373 basis because it alone priced turnkey civil (\$4.56M) and installation.
- **March 2026 contract, \$12.1M (gold):** supply only, with supervision. Civil, install, GSUs, station and interconnection moved to Eversource scope.
- **Equipment-only inside the contract:** \$10.64M, \$532/kWh, in line with published 2025-26 turnkey system indices (slides 9-10).
- **Rev 3 adds \$1.13M of escalation** to a price the contract fixes until Final Acceptance.

Three assumptions that, if wrong, move the number more than the reserve

1. Construction may be low, not high.

Rev 3 layers B+D+E are \$8.3M direct. Haugland's Class 3 base for overlapping scope is \$19.2M with a \$2.9M deduct ceiling. Nobody has mapped one to the other.

Settles it: a signed HEG-to-Rev 3 scope map before the PAF is drafted.

2. Siting is binary and carries no float.

ZBA determination, construction start and container delivery all land in Q1 2028 for a Dec 2028 ISD. Denial means an EFSB exemption at 18-24 months. The estimate carries \$250K for siting delay.

Settles it: Article 80 filing date, ZBA calendar, P6 float on the siting chain.

3. The number is Class 4 wearing Class 3 clothes.

30% engineering, unit-library construction with no bid, install labor as a percentage of a 2023 total, a 2.5 ft platform placeholder. EPAC will hear \$57.2M as a planning number.

Settles it: geotech, HMA, HEG map, crew build-up; then reclassify with a -15/+25% range.

Stress tests on our own numbers, and what history says about small utility batteries

Scenario	Loaded cost	Date	Covered by \$7.6M reserve?
Construction unit costs +25% on B+D+E+G	+\$4.0M	0	Yes, alone
HEG reconciliation at midpoint of the gap (+ \$5M direct)	+\$8.4M	0	No
Schedule +12 months (siting or long-lead)	~+\$5M	+12 mo	Partly
EFSB zoning exemption path	~+\$8M to + \$10M	+18-24 mo	No
Contractor bids +20% over estimate labor	+\$2.1M	0	Yes, alone
Combination: HEG midpoint + 12-month slip	~+\$13M (to ~\$70M)	+12 mo	No
Battery change orders at the LD cap (10%)	+\$1.2M	0	Yes

Untestable from the package: tariff or change-in-law pass-through on the Nomad price (Exhibit C terms), the AFUDC rate (assumed ~7% on CWIP), Station 496 outage premium (1.35 to 1.70 factor on affected hours), FEOC / ITC effect on net cost.

Base rate: approved to actual

Project	Uplift
BGE Fairhaven 2.5 MW/9.7 MWh	+64%
HECO Waena 40 MW/160 MWh	+37%
SaskPower Regina 20 MW	+31%
Eversource Provincetown	+9% to +40%
Hyde Park, Dec 2023 to Sep 2026	+30%
Hyde Park, Rev 0 to Rev 3 (5 wks)	+21%

Rev 3 carries 17.9% risk plus contingency. The exception case for beating the base rate rests on three anchors: a fixed battery price, a Siemens price on the skids, and Haugland engaged pre-construction. Together they cover about a third of the cost. The other two thirds have no such anchor.

Pre-mortem: June 2031, \$74M, fifteen months late. What happened?

The ZBA conditioned approval on a smaller footprint and new screening; the civil package went back through Haugland in the winter of 2028. Spring 2027 borings found debris deeper than the 2014 closure set showed; the platform went from 2.5 ft of fill to piles, and the reconciliation that had never been done was done as a change order. Nomad's twelve containers arrived on time in January 2028 and sat in a rented yard for nine months. Station 496 cutovers needed night outages priced at straight time. AFUDC alone added \$4M. The PAF had said Conceptual, -25/+50%. Everyone remembered \$57M.

Early-warning indicators

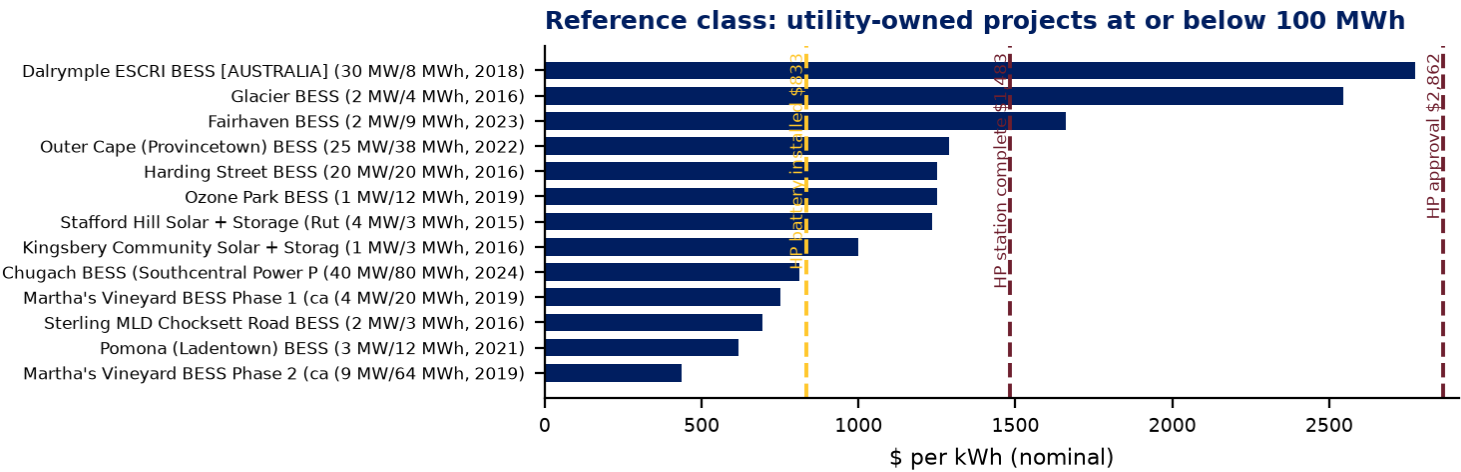
Indicator	Caught today?
Geotech report not issued by IFC-civil design freeze (mid-2027)	No: no P6 gate ties IFC to it
ZBA hearing date not set within 90 days of Article 80 filing	No: siting carries hours, not milestones
HEG reconciliation not closed before the PAF is drafted (Nov 2026)	Only if the CoE makes it a precondition
Nomad final-drawings milestone (13 Jul 2026) slipped	Partly: payments tracked, not downstream effect on GSU spec and civil
Station 496 outage plan not issued before construction pricing	No

What must close before this number goes to EPAC as full funding

Item	Type	Dollars at stake	Owner
Reconcile Haugland Class 3 base (\$19.2M, its scope) to Rev 3 layers B+D+E (\$8.3M direct). Red team expects it to move the base up, not down.	Scope boundary	+\$4M to +\$8M direct if half the gap is 21334 scope	Cost Estimating + PM + HEG
Siting chain has no float: ZBA determination Feb 2028, construction start Feb 2028, ISD Dec 2028; EFSB fallback 18-24 mo	Schedule	~\$5M per 12 months (AFUDC, escalation, PM, indirects)	Siting + PM
Present full funding at its real maturity: 30% engineering, no bid, placeholder platform = Class 4	Classification	Range is -25/+50% (\$43M-\$86M) until closed	Cost Estimating
Geotechnical investigation and NFPA 855 hazard mitigation analysis before civil freeze	Study	\$1.0M of deducts vs \$0.6M-\$3.5M exposure	Civil + Capital Projects Eng.
Confirm four written design-basis items (retaining wall, basin, duct bank, GSU count)	Design confirmation	\$1.47M HEG deduct ceiling (~\$0.8M net of GSUs already at 4)	Substation Engineering
Escalation on the fixed-price battery balance; check Exhibit C for tariff / change-in-law pass-through	Decision	\$1.13M	PM / Cost Estimating / Sourcing
Battery installation labor: replace 25%-of-16,801 mh with a crew build-up	Estimate	+/- \$0.5M (Rev 2 used 50%)	Cost Estimating + Nomad + HEG
Station 496 at \$363K per relay: compare to two recent EMA relay-replacement actuals; outage-window premium not priced	Benchmark	Unknown; \$1.8M testing line	P&C Engineering
Spot generation if the BESS misses Dec 2028 (SSF: 2-8 MW) was removed from scope; who carries it?	Scope / ownership	Not in project	System Planning
Re-base D-Line 24211 (2023 conceptual \$4.8M; HEG take-off larger)	Estimate	Unknown	Distribution Engineering

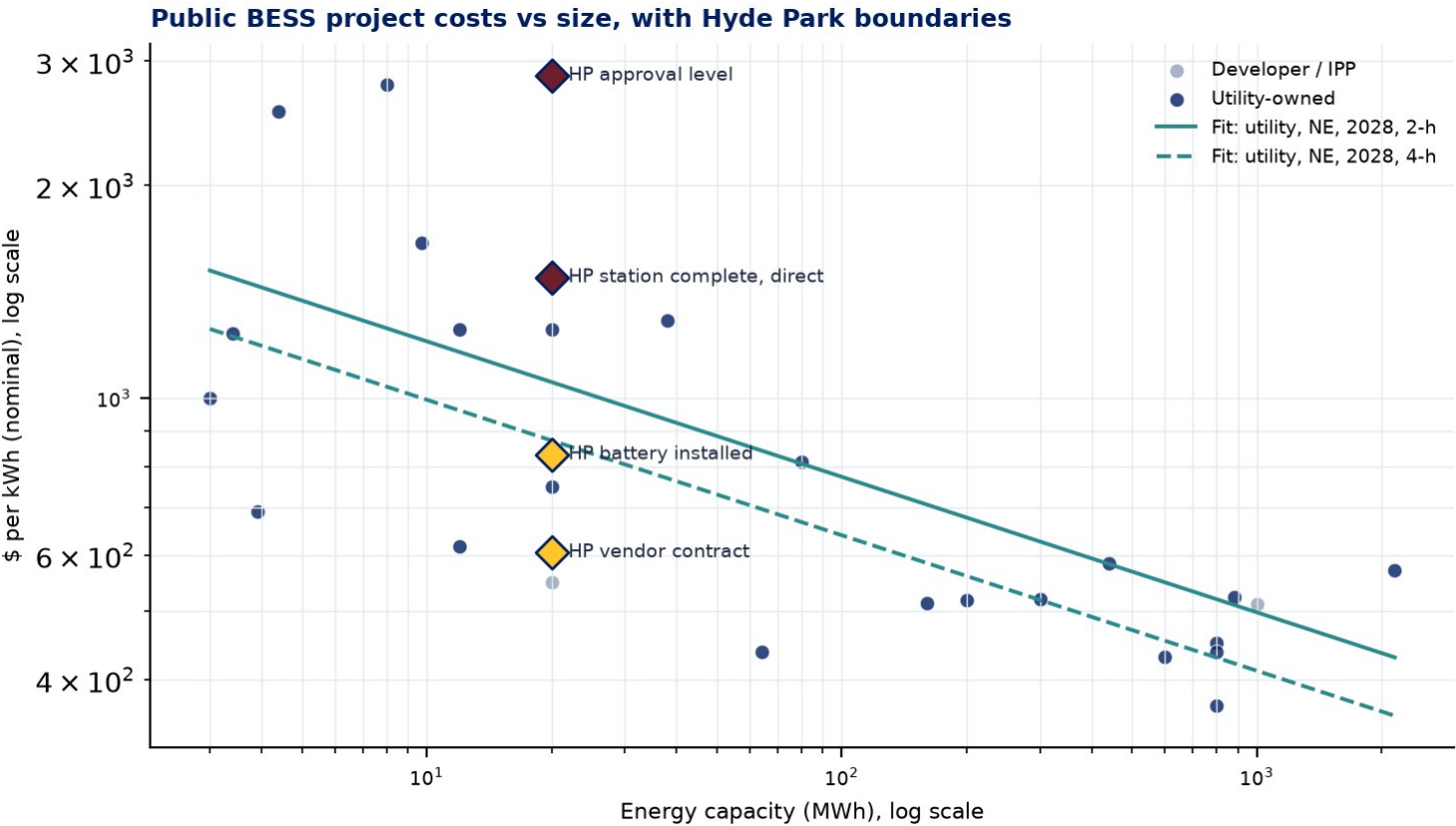
Defensible reductions (~\$3M: battery escalation, design-basis confirmations, material risk sized on non-battery material) are conditional on the Haugland reconciliation closing first.

Thirteen utility-owned projects at or below 100 MWh: where Hyde Park sits



- **Median** \$1,235/kWh; interquartile \$750 to \$1,289.
- **Hyde Park battery installed, \$833:** 38% percentile.
- **Station complete, direct, \$1,483:** 77% percentile, beside Fairhaven (\$1,659 actual, 2023), Chateaugay (\$1,490, 2023) and Provincetown (\$1,289, 2022).
- **Approval level, \$2,862:** above every member. The loaders (\$23.7M) and the two interconnection scopes are what the comparables do not carry.
- **Caveat:** figures retrieved from search excerpts of regulator and utility documents; verify at source before external use.

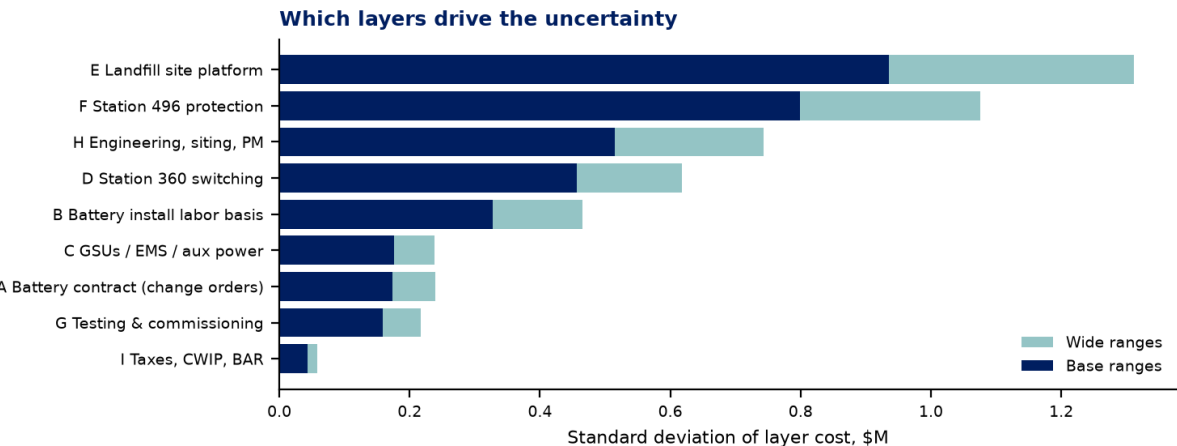
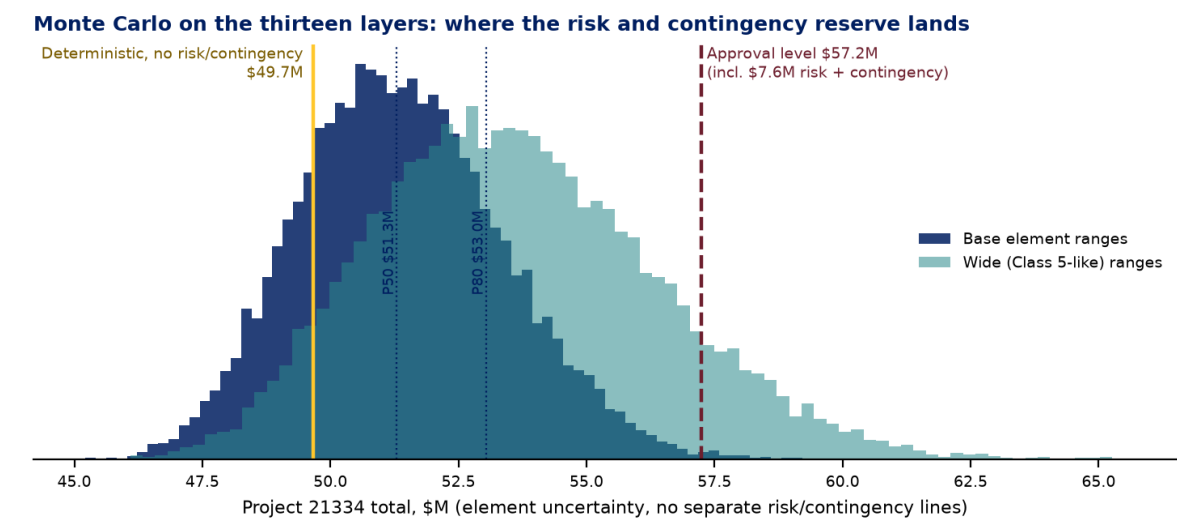
Size and duration explain two thirds of the spread in public costs; Hyde Park's direct cost is high-normal



Term	Coefficient	p
Size (ln MWh)	-0.17	0.012
Duration (ln h)	-0.29	0.016
Year	+0.015	0.68
R-squared	0.65	n = 26

- **20 MWh, 2-h, 2028 prediction:** \$1,116/kWh, 80% interval \$611 to \$2,037.
- **Hyde Park percentiles:** contract 10%, installed 27%, station direct 73%, approval 98%.
- **Use as a bracket, not a price.** Residual spread is a factor of 1.4; ownership and region terms are not significant at this n.

Monte Carlo on the layers: reserve covers the modeled ranges, thin against history



	Base ranges	Wide ranges
P50 total	\$51.3M	\$53.4M
P80 total	\$53.0M	\$55.9M
Reserve to reach P80	\$3.4M	\$6.2M
Carried risk + contingency	\$7.6M	\$7.6M
P(total > \$57.2M)	0%	10%

- Distributions** are PERT ranges on each of the nine Station 360 layers plus Station 496, indirects and AFUDC; risk register and contingency are not stacked on top (they fund the same outcomes).
- Estimate-to-actual uplift on comparables:** Fairhaven +64%, Provincetown +9 to +40%, Waena +37%, Regina +31%, Hyde Park Sep cycle +21%. Median about +31%; Rev 3 carries 17.9%.
- Next step:** replace these ranges with the project team's in a two-hour range-estimating session (AACE 41R-08) before full funding.

CONTRIBUTIONS

A \$12 million battery is a \$57 million project, and now every dollar between them has a name

- **Where the money sits.** 292 lines in thirteen layers, reconciled to the penny: battery contract 21%, station and interconnection 32% loaded, owner scope 19%, loaders 41%.
- **Unit cost at every boundary.** \$605/kWh vendor, \$833 installed, \$1,483 station direct, \$2,862 project. Each benchmark read against the row that shares its definition.
- **How it changed.** The battery got cheaper; the project got real. \$43.9M to \$57.2M, five in-service dates, documented from the ESF revisions and the email record.
- **Can it be defended.** Inside the reference class on direct cost; three kill shots, seven stress tests, a pre-mortem, and ten reconciliations with owners and dollars.

Decisions requested

- Adopt the layered view (battery, station, interconnection, owner scope, loaders) as the standard presentation of this project, internally and to regulators.
- Make the Haugland-to-Rev 3 reconciliation, the geotech and the hazard mitigation analysis preconditions of the November PAF; present at Class 4 until they close.
- Assign owners to the siting-chain float and the five early-warning indicators.